

SAT-to-GS

Constraint: SAT Ops Cadence

Cost Function: Each GS Revisit

Earth Observation SAT Image Capture Scheduling

- Get image data from ground by any available satellite
- Minimize revisit time for each ground operation point

- Worldwide Distributed Observation
- Customize τ (Operation Cadence) for each SAT / G to Prioritize Ground point

6G SAT Data Downlink/Uplink Scheduling (With ISL)

- Uplink/Downlink data packet from GS to any satellite
- Minimize time gap between each transmission
- Assume every satellite is connected each other by ISL

- Regional Lattice Points Communication
- Place Multiple Communication Payloads in single SAT (Multi-Beamforming)